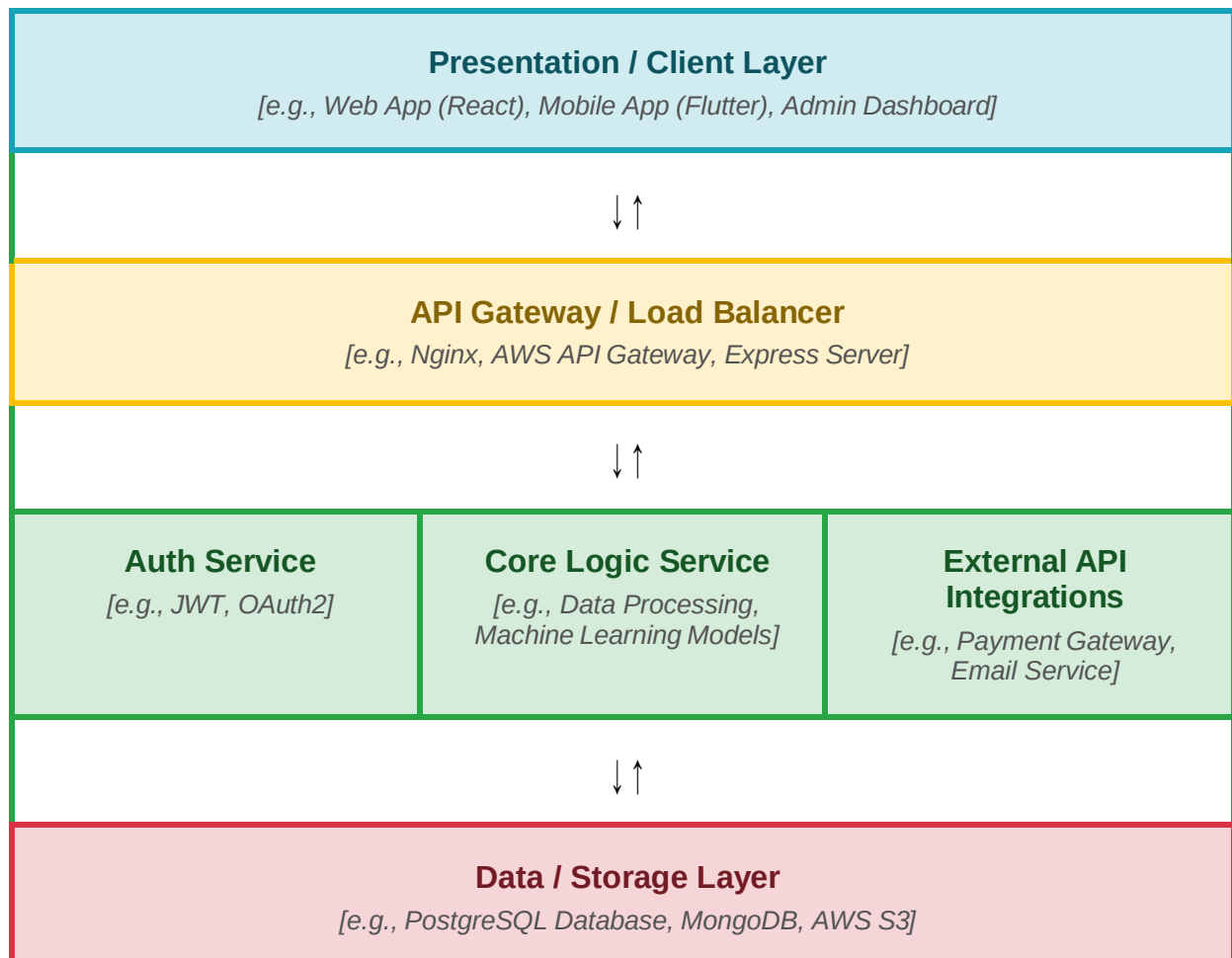


Solution Architecture

Date	24 June 2026
Team ID	
Project Name	EduGenie
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provided a high-level architectural overview of the EduGenie platform. The architecture diagram illustrates the interaction between the Presentation Layer, Application Layer, and Data Layer, showing how student requests are processed, AI-generated educational content is delivered, and learning data is stored and managed. Complete the architectural diagram by filling in the relevant technologies and components used in each layer of the EduGenie system.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>User Interface (HTML/CSS/JS)</i>	Provides a responsive user interface where students interact with the AI Q&A, explainers, quizzes, and learning paths.	HTML5, CSS3, Vanilla JavaScript, Jinja2
<i>FastAPI Server</i>	Handles routing, application configuration, request validation, authentication dependencies, and serves static files.	FastAPI (Uvicorn router)
<i>EduGenie AI Engine</i>	Handles core business logic, including auth_service, rag_service (RAG document loading and processing), and AI service integrations (Gemini, LaMini).	Python, google-generativeai, PyPDF2, python-docx
<i>JSON-Based Data Storage</i>	Stores user information, quizzes, summaries, learning paths, and application records in structured JSON files.	JSON, Python